

From Sensable to Sensible Spaces: Enhancing the Sensibility of a Home Office using Stress-Aware Deep Reinforcement Learning in Virtual Environments

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Figure 1: My home office searching together with me for inspirational moments

ABSTRACT

Being confronted with a dulling perception and challenges to be creative when working at home, I imagined my home office to sense my state of arousal and search together with me for inspirational moments by infusing its familiar appearance with distortion.

In two prototypical human-home office interaction systems, we use Virtual Reality (VR), Deep Reinforcement Learning (DRL) and Galvanic Skin Response (GSR) to enhance a home office's sensibility towards its user's level of arousal as well as to enlarge its textural action space.

Although physiological feedback in machine learning faces low learning rates, the resulting interaction offers a fresh perspective on our human-home office relation.

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CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; • **Computing methodologies** → *Multi-agent reinforcement learning*; • **Applied computing** → *Media arts*.

KEYWORDS

sensible space, generative aesthetics, stress and affect detection, virtual environments, reinforcement learning, galvanic skin response, ml-agents

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1 INTRODUCTION

Welcome to my home office.

I have spent the last two years in here. Before the pandemic, my home was my base for relaxation and time off - familiar and with low arousal.

As an artist, creativity and inspiration are essential to my work. Usually, my sources of inspiration are arousals and unplanned impressions throughout my day.

When during the pandemic I suddenly had to work from home, I had a hard time being creative. I needed to break out, but of course, without actually leaving the room.

I 3D-scanned my home office. As you can see, the resulting virtual reality is not perfect at all. Just look at the blob of flowers on my table or my notebook without its screen (Figure 1). These distortions from reality already triggered something in me. So I decided to take it even further.

I started to experiment with tracking my arousal through skin conductivity. And I hooked this real-time data up with a machine learning model that then learns to transform my virtual home office.

By doing so, we aim to research the inspirational potentials of enhancing a home office's sensibility towards its user. This work is intentionally set at the intersection of various research fields as we hope to contribute to bridging the discourse in those.

2 THEORETICAL BACKGROUND

Within any creative job, feeling inspired is vital. Inspiration, the moment of being stimulated to be creative, can be regarded as both a trait and a state [23]. As a trait, it might vary from individual to individual. As a state, it can vary within an individual depending on different variables. One of these variables is the environment. Studies have shown that, e.g. office design has an impact on employee's creativity [25][4]. Within one physical home office, the action space for aesthetic stimuli is limited but could be extended through Augmented Reality or Virtual Reality (VR). Still, as aesthetic experience is a highly individual and dynamic process [16], even with a larger space of aesthetic possibilities reaching and reproducing a specific state of mind can be very challenging [7].

Galvanic Skin Response (GSR), also known as Electrodermal Activity (EDA), refers to the increase in skin conductance or the decrease in skin resistance due to sweat gland secretion measured on the palms of the hands or the feet. These alterations are evidence of the activation of the sympathetic nervous system reflecting unconscious cognitive, attention, and emotional changes such as arousal and stress [19]. Although individuals vary significantly in the way they perceive and react to the same stressor, long term exposure causes chronic stress, as well as mental and physical health issues [8]. Interestingly enough, stress may also stem from pleasant events, occurring positive sentiments, or performance increase [17]. Because of the preceding, predicting stress in places such as work using wearable, unobtrusive sensors, including GSR, and building personalized stress detection algorithms have received growing attention [13][9][24].

When exploring options of extending the action space of a VR space, texture seemed like a natural approach offering a wide range of possibilities. As "familiarity" and "fluidity" are two influential components in aesthetic appreciation [15], a certain similarity to

known textures and a fluid transition between those seemed important. Several research groups have created large databases of textures [10][1]. Learning and mapping a large space of images is one of the features of Generative Adversarial Networks (GAN) [12][3]. When having a vast GAN latent space navigating it in a meaningful way is essential. Here Deep Reinforcement Machine Learning (DRL) and its ability to react upon observations in real-time come into play [6]. Additionally, Unity has provided the "ml-agents" framework, which allows an accessible way of implementing DRL in virtual environments [11].

In line with Emanuele Coccia's concept of Sensible Life, this paper is written based on a differentiation between object space, images and sensible subjects [2]. According to this ontology, objects are not directly perceivable but become sensible through so-called images. The term image is here understood as a sensible sphere containing all sensible information, not limited to the perception capabilities of a single entity/species. On the other hand, sensible subjects are defined as entities able to sense those images and change their own appearance. In line with this metaphysical concept, my home office was present in the object space and provided a sensible image. However, it was not able to act as a sensible subject, sense my image and adapt its own.

3 METHOD

3.0.1 Setting. The motivation of this paper was the home office situation of myself. I am 30 years old and live with my partner in a 47m² apartment. My home office space is the living room. It has an area of 4 x 5m and a height of 3.5m. It has two high windows towards the street, one door to the sleeping room and one to the entrance space. The work I am doing in my home office is partially my artistic practice and partly academic research. During the pandemic lockdowns, I spent most of my non-sleeping time in this room: breakfast, work, lunch, work, dinner, work and some free time.

3.0.2 Virtual Environment. As a preparation for the prototypical system presented in this paper, the home office space was 3d-scanned with the lidar sensor on an iPad Pro. The result was a coloured point cloud with 286,203 points. The home office 3d scan was then imported to Unity and enhanced with a custom shader script and shader graph. The shader script adds the ability to transition smoothly from one texture to the next one. The shader graph creates a continuous fluid vertex displacement movement relative to the texture RGB values. As an output device, we use the Valve Index VR headset.

3.0.3 Textural Action Space. The texture action space is framed and mapped by the latent space of a pre-trained StyleGAN2-ADA network [12]. This StyleGAN2 was pre-trained on the Describable Textures Dataset (DTD) [1]. An amount of 512 vectors between -3.0 and 3.0 define the latent space. This StyleGAN2 pipeline runs in a Python environment and waits for latent vectors sent by the later-described DRL agent via UDP. For every latent vector set received, the StyleGAN2 generates the associated texture and streams its RGB values back to Unity.

3.0.4 Stress and Affect Detection. The GSR data is collected with an Empatica E4 wristwatch at a sampling rate of 4Hz. The sensor data

is streamed to the VR host PC through the E4 streaming software. The raw GSR data is then processed with a personalized stress classification pipeline. The classification model was trained on GSR data from Wearable Stress and Affect Detection (WESAD)[20] benchmark dataset along with data collected in a stress-inducing experiment at four sessions. Each of these sessions was conducted in VR (Figure 2) [22]. The experiment protocol included a neutral reading to obtain a baseline, a short horror movie to induce stress [21], and a meditation task to relieve stress. After each task, a subjective stress evaluation (1 - 'not stressed' to 10 - 'very stressed') was assigned to serve as ground truth for perceived stress. The study protocol of the WESAD dataset also involves baseline recording, stress-inducing and meditation tasks. Inspired by Nkurikiyeyezu et al.[18], we aimed at deploying a personalized stress detection model utilizing benchmark data as generic samples and data from VR sessions as calibration samples. More specifically, the training dataset comprises GSR data samples from all 15 WESAD subjects along with a few calibration GSR data samples from our subject at VR sessions. The latter conveys the unique person-specific biometrics that facilitates subject-independent stress recognition. The remaining samples are used as the test set. Evaluation of the presented framework yielded an accuracy of 91.2% in predicting between calm and stressful moments.



Figure 2: Screenshot of the VR stress-inducing experiment

3.0.5 Online GSR Processing. In preprocessing, a median filter with a 5-second kernel size is employed to remove noise such as motion artefacts and signals are transformed to a -1 to 1 range to account for individual and intravidual differences. In the training phase, a Support Vector Machine with Radial Basis Function kernel was selected as a classifier, and a five-fold grid search over the C, gamma parameters and the number of features was performed. The feature vector extracted as input to the classifier consists of 14 features, namely, mean, standard deviation, max and min of the EDA signal, max value of peaks of the EDA signal, average of total Skin Conductance Response (SCR), max of SCR, number of peaks in the SCR, max of SCR amplitude, the average of SCR peak amplitude, the average of total Skin Conductance Level (SCL), the difference between maximum SCL and minimum SCL, skewness and kurtosis. In real-time, the GSR data is streamed using LSL [14], stored in a 60-second buffer and updated every 30 seconds. Preprocessing steps are the same as in the training phase. For classification, the

trained stress detection model is loaded. Consequently, a binary decision upon a stressed or relaxed state is streamed to the DRL agent. Additionally, we also stream the current raw GSR feature vector to observe the current skin state in more detail.

3.0.6 Online Reinforcement Learning. With the use of ml-agents, the virtual home office is now ready to become a DRL agent able to observe, act, and learn [11]. Both experiments presented in this paper were conducted with the “Proximal Policy Optimization (PPO)” reinforcement learning algorithm. In every action step, the home office DRL agent first observes the current StyleGAN2 latent vectors, the last arousal prediction, the current GSR vectors and the current target arousal (Figure 3). Based on these observations, it then decides upon either increasing or decreasing each of the 512 vectors in a step size of 0.2. Afterwards, it sends the new latent vectors to the StyleGAN2 pipeline, gets back the associated texture and starts to transition towards it. Once a new arousal prediction is ready, the home office DRL agent calculates its reward by subtracting the predicted arousal value from the targeted arousal value. The targeted arousal value can be defined as a constant or curve. When expressed as a curve, e.g. sine wave, the step specific target arousal is calculated by calculating the curve value associated with the current step count.

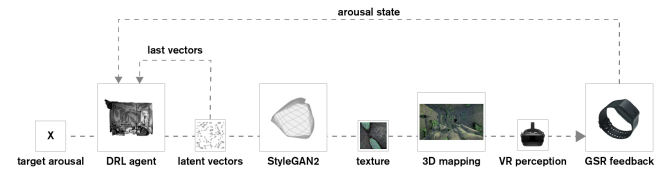


Figure 3: Adapted DRL Setup with StyleGAN2 latent space navigation, VR output and GSR feedback

3.0.7 Interpretability. The Home Office DRL agent always initializes with the same default latent vectors. Therefore, the initial StyleGAN2 texture is always the same. Additionally, as the DRL agent navigates the GAN latent space, the specific textures do not indicate the user’s arousal feedback. Also, the frequency of transitions is not related to the user feedback. Only if the DRL agent is trained for a sophisticated number of samples the amount of colour and pattern change could be understood as a hint of how close targeted arousal and predicted arousal is.

3.0.8 Data Privacy. Within the scope of this method, GSR data is only saved during the collection of the SVM training data. The data collection happened consensually with the test person - being one of the authors of this paper - and the CERTH Ethics Committee. The collected GSR data was saved anonymously only for research purposes. Once the SVM is trained, no further GSR data is saved, only processed. Additionally, the whole system architecture is designed to run all processes locally.

4 PROJECTS

Over one year, we developed two prototypical interaction systems at the intersection of aesthetic research, VR, machine learning,

creativity and decentral work - each exploring a different strategy for approaching the phenomenon of inspiration.

- “Zoom Out A” is a reactive VR experience that searches with you a moment of inspiration by targeting a constant value of arousal.
- “Zoom Out B” is a reactive VR experience that searches with you a moment of inspiration by targeting a narration of arousal.

Both VR experiences allow for controller-free walking around and are realized without sound. Viewers wearing the VR headset are free to walk, sit and stand as preferred. The virtual home office scale and rotation is synched to the physical home office.

4.1 Zoom Out A

“Zoom Out A” is a VR artwork that uses DRL and GSR to sense and learn with the interactant’s level of arousal. In “Zoom Out A”, the home office targets a constant arousal value. Respectively the home office tries to reach and hold the defined arousal response.

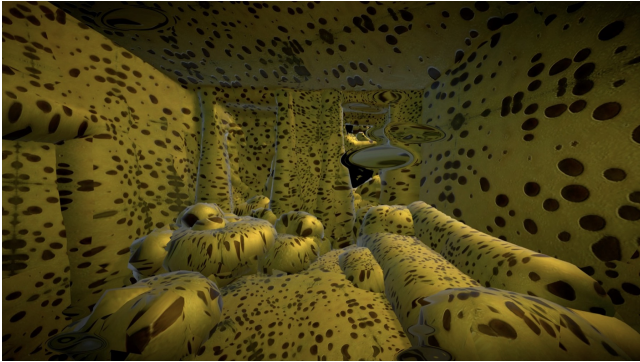


Figure 4: Screenshot of the VR experience Zoom Out A

4.1.1 Design Rationale. As already mentioned in the Theoretical Background section, we tried to build the experience design around specific features of aesthetic perception found to increase the likelihood of appealing appearance. One of them being “familiarity”. Familiarity can be best understood as an in the current context relatable experience, a non-alien experience. This led to the decision to train the StyleGAN on a natural texture database. By not separating the point cloud into known objects with independent textures (e.g. chair, couch, table) but treating it as an immersive canvas, we tried to integrate a second feature known as “simplicity”. The third feature we focused on is “fluidity”. For us, it was essential to have the possibility to adapt texture dynamically without a disturbing moment of transition. Therefore we implemented two dynamics. One is a constant smooth transition from one texture to the next, and the other is a constantly changing vertex displacement. Besides fulfilling a demand for fluidity, we also expect this constant movement to be of narrational importance. It avoids peak points and keeps the curiosity up while still in expected borders.

4.1.2 Procedure. The GSR wristwatch is put on first. Afterwards, the GSR streaming software is linked with the wristwatch. Next, StyleGAN2, DRL and Unity VR environments are initiated. Once

the VR setup with DRL, StyleGAN2 and GSR streaming is running, the user has to put on the VR headset, and the search for a moment of inspiration begins.

4.1.3 Specific learnings. We recommend a double-blind selection of the arousal target mode and unaffected test subjects for future studies in this direction. Being my own test subject, I knew that the surrounding space targets a constant arousal value. Targeting constant arousal is against my belief in an inspirational arousal narration, so I had difficulty freeing my mind. Nevertheless, for harvesting subjective results, it was of interest to get in contact with a sensing space following the agenda of pushing the arousal to a constant level. Compared to Zoom Out B, it led to a certain feeling of resistance and a need for taking my arousal and inspiration back under my control.

We started to wear the wristwatch 10-15 minutes ahead of the experiment to allow the wristwatch to get on body temperature and have more consistent measurements. Besides that, we would recommend not using VR hand controllers as those reduce the heat transmission of your palms, and the temperature of your hands could potentially affect the GSR feedback.

4.2 Zoom Out B

“Zoom Out B” also is a VR artwork that uses DRL and GSR to sense and learn with the interactant’s level of arousal. In “Zoom Out B”, the home office targets an arousal narration defined as a curve function. Respectively, the home office tries to achieve an arousal response in synch with the narration curve. Therefore, we expect Zoom Out B to act more sensible to individually definable arousal narrations when trained substantially.



Figure 5: Screenshot of the VR experience Zoom Out B

4.2.1 Design Rationale. For this experiment, the VR design and the initialized latent vectors at the first frame are the same as in Zoom Out A.

4.2.2 Procedure. Also, the procedure is the same as mentioned above. The only difference is that the viewer can define the narrational arousal curve ahead of the experience.

4.2.3 Specific learnings. Interacting with a space targeting an arousal narration without having feedback where on the curve I currently am subjectively lowered the barrier to open myself to

the process. Compared to Zoom Out A, it felt easier to focus on the body's awareness of its arousal and absorb the visual immersion. It could be interesting to look into the effects of different arousal curves and rhythms on the individual inspiration for further research.

5 RESULTS AND LEARNINGS

We did not conduct a representative study about this home office experiment. Still, we do have some results we want to share. Please read those as subjective reflections by the test subject.

From an objective point of view, the learning rate was remarkably low. Apparently, training DRL with physiological feedback takes weeks to achieve computational "meaningful" results. Subjectively, the feeling of being sensed by my home office and knowing its intention to search with me for inspirational moments already had a positive effect on my creativity. I would describe it as a feeling of another entity caring about me and knowing that I have support when searching for inspiration. This effect could be related to a similar phenomenon in work and organizational psychology, where employee motivation is already rising if managers change the workplace atmosphere - no matter the specific action [5].

Having a space with a known agenda and an action space visually surrounding myself while still graspable as a whole was also a significant part of the experience. We certainly are used to several spaces with agendas and action spaces. Still, their action space is usually limited to specific, less immersive actions, e.g. an automatic sunshade or a heating system.

One effect that stayed even when putting the VR headset off was the re-notation of the space. Before the first VR home office experiment, I was in my flat for around 14 days and no longer perceived any detail. If you had asked me which colour I connect with my home office, my answer would have been grey. Seeing this very same space first translated to a point cloud, then moving and in new textures changed my perception of it. I again perceived the colours and details, but besides that, I now also saw it as a canvas. The first effect could be named as regain of attention towards my home office space. The second effect could be understood as an aftereffect of a horizon-broadening experience.

Besides that, I felt that my body awareness of arousals and their effects on my inspiration increased through dedicating myself to these projects. Not finding the "inspirational moment" through technological means perhaps has provided me with the space to learn how to sense it.

It further puts light on the topic of how we could increase the work environment quality of home office workers in general and if inspiration sometimes might be experienced in a moment of muse rather than under time pressure.

6 DISCUSSION

We do not see this work applied as a tool for creativity in the standard Home Office context but as a method of researching the phenomenon of inspiration. We are aware that our prototypical setup does not consider the diversity of social, spatial, and technological home office situations. We purposely decided against broader user research favouring a more intense focus on a single user and a single environment, partially due to the pandemic, partly

because it allowed a much more qualitative and fruitful exchange throughout the process. We know that our chosen research method for this work primarily builds on subjective reflections. We, therefore, certainly see it as biased but hope to be still able to enrich the discourse about creativity and decentral work.

If this experiment is redone, we suggest experimenting with different reinforcement learning algorithms that might suit low sample learning better than the Proximal Policy Optimization (PPO) we used in this paper. We would also hope to inspire research into machine learning built for learning, unlearning and relearning together with humans.

For further research, we would emphasize a search for collective inspirational moments in virtual decentral work and the conductance of comparable physical experiments by materializing the presented method with perhaps real-time robotic control.

Overall, we hope to contribute to a transdisciplinary discourse about getting inspired within borders in times of borderless work.

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